Knocked Unconscious: Thematic Change in Popular Hip Hop Songs

Deborah Olorunișola

Yale University

Yale

Introduction

By the time the 2 Live Crew's sexually-explicit album *As Nasty as They Wanna Be* was declared legally obscene, hip hop was already known for its controversial themes. There is no question of the genre's capacity to generate controversy—its themes range from raunchy descriptions of sex to commentary on mass incarceration—but the source of its themes are up for debate. Many fans of the genre look to the 1990s as a concrete moment for the thematic change in the genre, arguing that it was strategically shifted away from "socially conscious" themes by large record labels, while others argue that the genre has always been full of "pureile" content. This begs the question: **Has rap music become less socially-conscious?**. Prior research approached an understanding of this change, but lacked a framework to analyze how "introspection" functions in the genre and focused only on 1979 to 1995. This project updates prior work and performs this analysis at scale using updated machine-learning methods.

PARENTAL ADVISORY EXPLICIT CONTENT

Data

The first source is the Billboard "Hot Rap Songs" chart from March 11th, 1989 until December 31st, 2024 with 4,598 unique songs. This data is available via the `billboard.py` API. The second source is *Ego Trip's Book of Rap Lists*, which supplements the missing lyrics data between 1979 to 1988 with 397 unique songs. Songs from these refrence books are available via Genius.com. Song lyrics for all the songs (n = 4,994) come from Genius.com via the `LyricsGenius` API.

Methods

Multilabel Classification

Define a dataset $\mathcal{X}$ with points $\mathbf{x} \in \mathbb{R}^d$. The goal is to find a mapping $\mathcal{X} \to \mathcal{Y}$ for l labels. This mapping is unknown, so we use machine-learning to train a classifier using labeled training data $\mathcal{D} = \{(\mathbf{x}_1, \mathbf{Y}_1), \cdots, (\mathbf{x}_N, \mathbf{Y}_N)\}$, where $\mathbf{Y}_i$ is the specific set of labels (one-hot encoding) corresponding to observation $\mathbf{x}_i$. The text embeddings representing each $\mathbf{x}$ were from a pre-trained BERT model (`Lyrics-BERT`).

Classifiers Employed

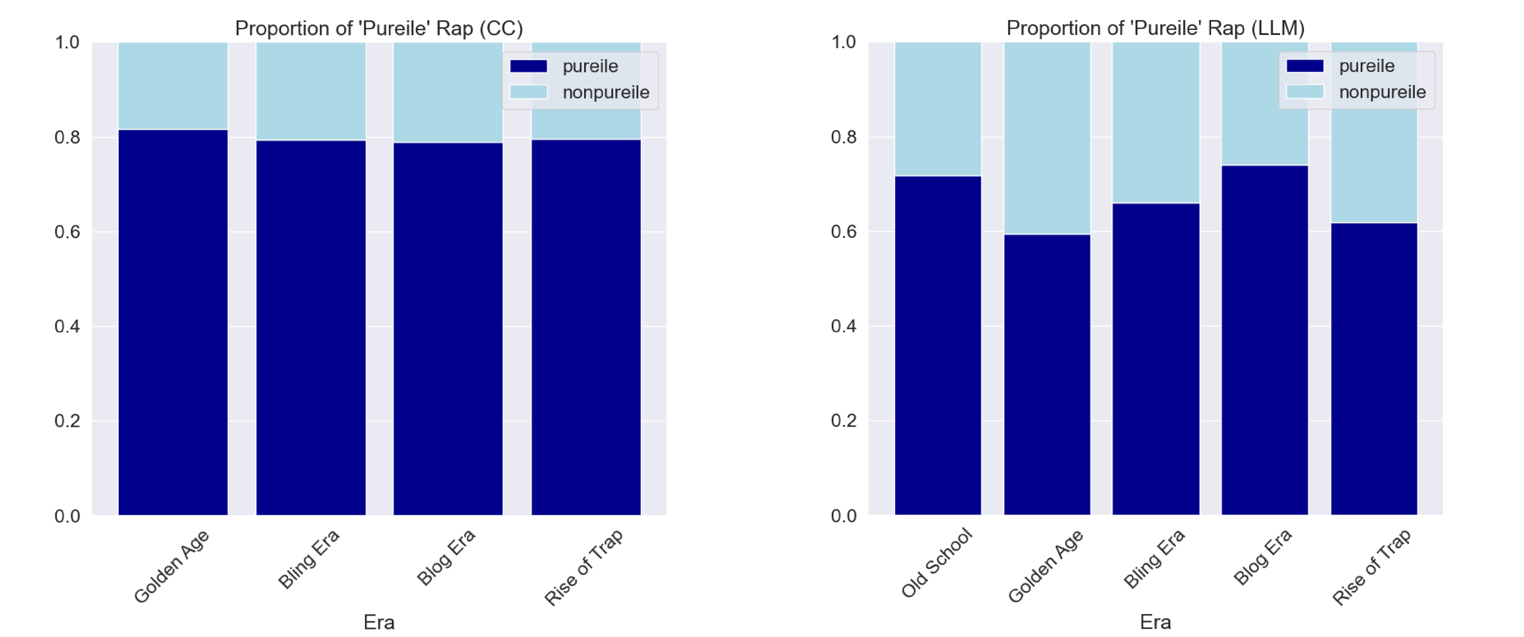
1. Binary Relevance (BR)
2. Label Powerset (LP)
3. Classifier Chain (CC) with imprecise correlation ordering
4. Multilabel K-Nearest Neighbors (MlKNN)
5. LLaMa 3.2 (LLM)

Key Findings

1. There appears to be little difference in the proportion of rap themes over the years (Fig. 1, 3).
2. MLC methods picked up on thematic spikes aligning with the Bling and Don Eras of hip hop, suggesting their utility for classification on short text in high dimensions (Fig. 2).
3. While the most prevalent theme in rap is still unclear, the genre appears heavily associated with braggadocio, street culture, sex and partying.
4. According to both models, socially conscious and introspective rap have never made up *most* of the popular songs in an era.

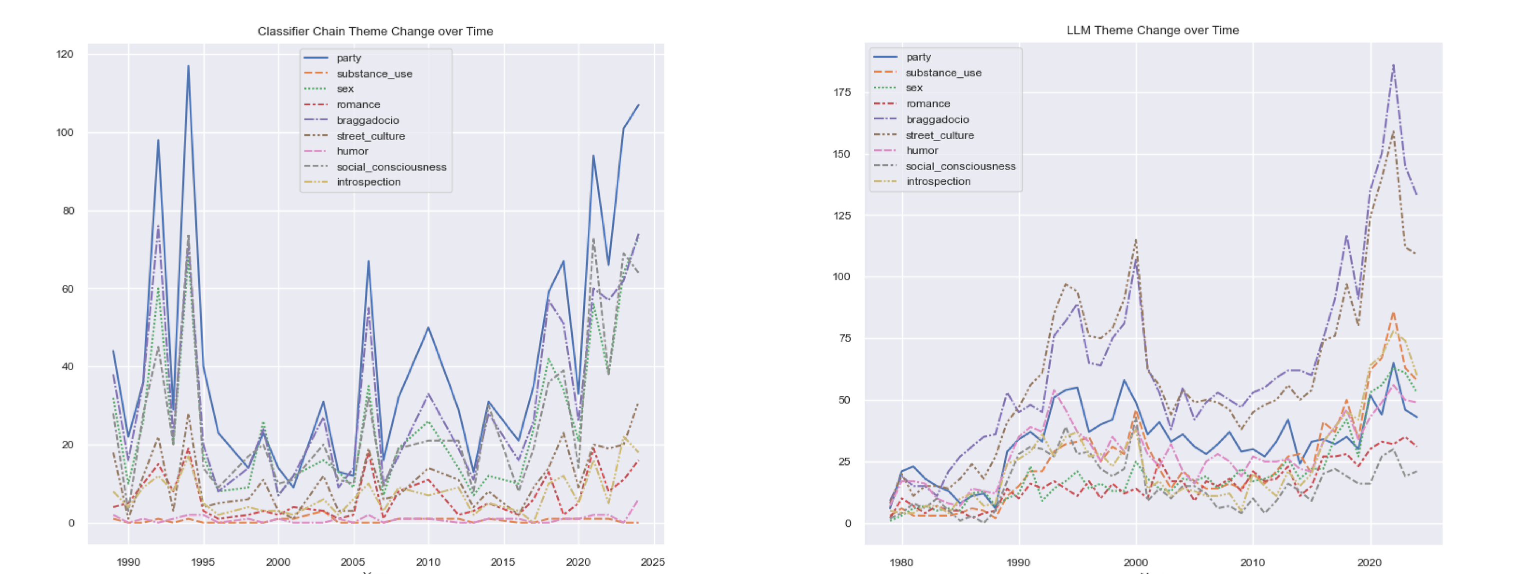
Results

Figure 1. **Differences in the prevalence of pureile rap, according to the top-performing models.** Non-pureile rap is defined as any song with themes of social-consciousness or introspection.



a. $\text{CC}_{\text{Impcorr}}$ Thematic Changes b. LLM_{drop} Thematic Changes

Figure 2. **Yearly changes in theme, according to the top-performing models.** The CC model suggests an increase in the proportion of braggadocious songs from the Bling Era, aligning with prior research.



a. $\text{CC}_{\text{Impcorr}}$ Thematic Changes b. LLM_{drop} Thematic Changes

Figure 3. **Distribution of Themes from $\text{CC}_{\text{impcorr}}$.** According to this method, rap lyrics are related to partying and sex.

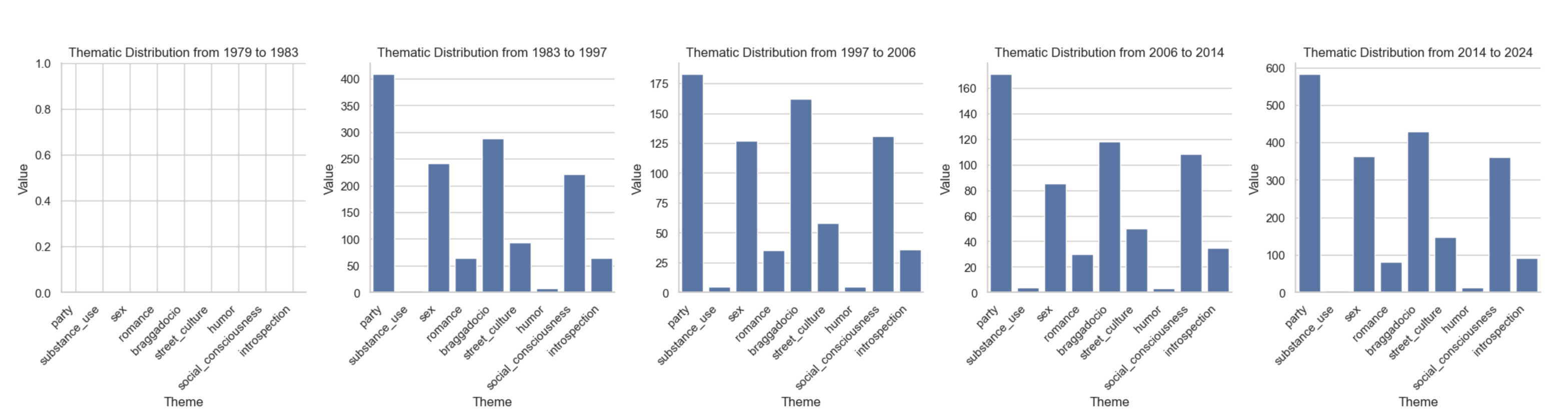
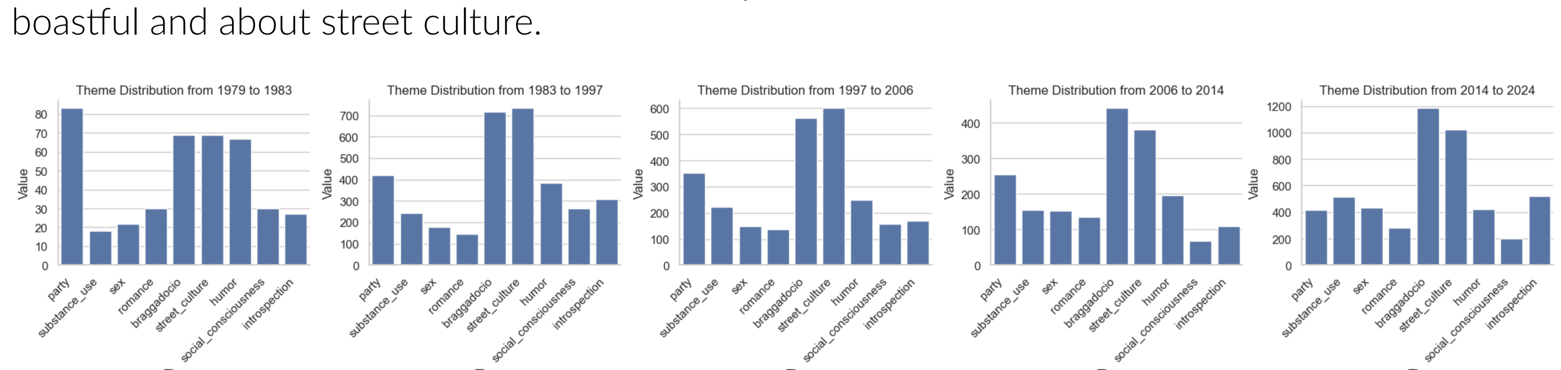


Figure 4. **Distribution of Themes from LLM_{drop}.** According to this model, the rap ethos is mostly boastful and about street culture.



Model Performance

Table 1. **Performance of each algorithm per each evaluation metric.** The precision and accuracy of $\text{CC}_{\text{Impcorr}}$ provide more confidence that the learned labels would align more with what a human might say. LLM_{drop} has the benefit of high recall (and thus, high F_1), meaning it is very sensitive to positive samples, getting nearly half of all the true positives in the test set.

Metric	BR	LP	$\text{CC}_{\text{Ensemble}}$	$\text{CC}_{\text{Impcorr}}$	MlKNN	LLM	LLM_{drop}
Hamming Loss	**0.198** (1)	0.274 (5)	0.216 (3)	0.211 (2)	0.224 (4)	0.384 (6)	0.385 (7)
Accuracy	0.005 (5)	0.024 (3)	**0.033** (1)	0.029 (2)	0.019 (4)	0 (6)	0 (6)
Micro F1	0.005 (7)	0.137 (3)	0.104 (4)	0.090 (6)	0.092 (5)	**0.283** (1)	0.251 (2)
Macro F1	0.004 (7)	0.076 (3)	0.065 (4)	0.058 (5)	0.042 (6)	**0.255** (1)	0.211 (2)
Precision	0.100 (7)	0.186 (6)	0.272 (2)	**0.278** (1)	0.218 (3)	0.208 (4)	0.188 (5)
Recall	0.003 (7)	0.109 (3)	0.065 (4)	0.054 (6)	0.062 (5)	**0.443** (1)	0.377 (2)
Average Rank	5.67	3.83	**3**	3.67	4.5	3.17	4

Model Selection

Classifier Chain

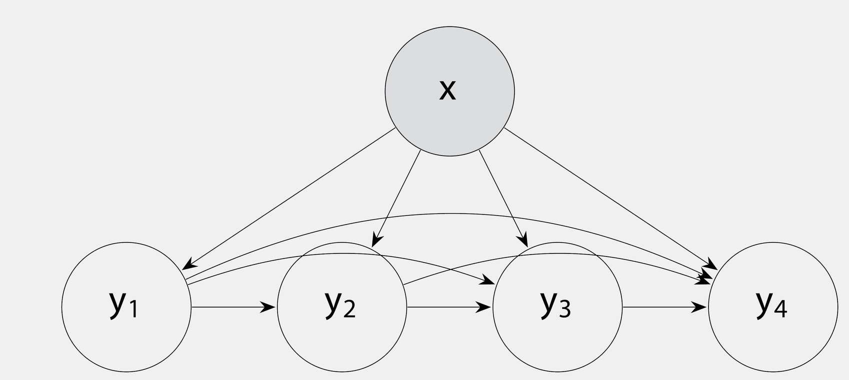


Figure 5. Modeling the relationships between labels in a classifier chain.

Classifier chains turn the multilabel classification problem into a series of binary relevance problems, but models the relationship between labels. To decide on the optimal label ordering, I adapted the Imprecise Correlation (`Impcorr`) method. Given the distribution of labels in the training data, this algorithm calculates

$$ISU(y_j, y_k) = \frac{2IIG(y_j, y_k)}{S^*(P(y_j)) + S^*(P(y_k))}$$

and finds the next most likely label.

Large Language Model

LLaMa 3.2 is an open source large language model. To adjust the model for these research purposes, I used Few-Shot and Chain-of-Thought prompting. The temperature of the model was 0.75 and the system message is available at the GitHub for this project.

References

[1] Michael Eric Dyson. *Know What I Mean? : Reflections on Hip-Hop*. Basic Books, New York, UNITED STATES, 2007.

[2] Jennifer C. Lena. Social context and musical content of rap music, 1979-1995. *Social Forces*, 85(1):479–495, 2006.

[3] Serafin. Moral-García, Javier G. Castellano, Carlos J. Mantas, and Joaquín Abellán. A new label ordering method in classifier chains based on imprecise probabilities. *Neurocomputing*, 487:34–45, 2022.

[4] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. Llama: Open and efficient foundation language models, 2023.